

Federated Learning for ICU Mortality Prediction on Heterogeneous Clinical Data

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Abstract—Privacy regulations such as HIPAA and GDPR restrict multi-center pooling of raw electronic health records, while institutional datasets are often insufficient for robust clinical predictive modeling. This research evaluates privacy-preserving and Byzantine-robust Federated Learning (FL) for Intensive Care Unit (ICU) mortality prediction under realistic Non-IID conditions. Using the MIMIC-IV dataset (65,273 hospitalizations, 31 clinical features) partitioned into seven distinct care units, we benchmark server aggregation schemes (FedAvg, FedProx, coordinate-wise Median, Krum, and validation-calibrated FedF2) alongside local differential privacy (DP-SGD, $\epsilon = 4.36$, $\delta = 10^{-5}$). The centralized logistic regression baseline attains 0.8920 AUROC (85.2% recall at $\tau = 0.39$). In the federated setting, Platt-scaled logistic regression preserves discriminative parity at 0.8784 AUROC ($< 1.5\%$ loss), while exploratory federated XGBoost via ensemble voting yields 0.9155 AUROC. Under a 3-of-7 adversarial client attack, Median and Krum maintain robust AUROCs of ≥ 0.8594 and ≈ 0.851 , whereas FedAvg collapses to 0.5011. Formally private DP-SGD achieves 0.8477 AUROC. Cross-dataset federated replication on eICU-CRD across seven independent hospitals achieves 0.8349 AUROC (1.12% loss vs. centralized). These findings show that federated learning is clinically feasible across hospital networks, provided Byzantine-robust aggregation is deployed when client trust cannot be guaranteed and decision thresholds are calibrated to clinical operating requirements.

Index Terms—Federated Learning, Healthcare Machine Learning, Differential Privacy, Non-IID Data, Byzantine Robustness

I. INTRODUCTION

The healthcare industry has developed numerous mechanisms for collecting medical data in clinical institutions at an unprecedented rate, including electronic health records, radiology image storage, and laboratory test data. However, virtually no exchange of this data takes place between different healthcare organizations, primarily because existing legislation such as the Health Insurance Portability and Accountability Act (HIPAA) in the US and the General Data Protection Regulation (GDPR) in the European Union prohibits sharing patient data outside institutional boundaries [1], [2]. Additionally, there has been a significant rise in ransomware attacks on hospital IT systems, increasing risks in centralized data storage [3]. In effect, hospitals have their own islands of data. Machine learning algorithms that require extensive datasets to train effectively cannot access them [4].

Federated Learning (FL) was proposed precisely to work around this constraint [5]. The idea is straightforward: data

never moves. Instead, every participating hospital trains a local copy of the model and transmits only the resulting weight updates to a central server, which fuses them into a shared global model [6]. No individual patient record ever leaves its home institution, supporting compliance with HIPAA and GDPR privacy principles by retaining raw EHR data on local hospital premises. Researchers have already tested variants of this approach on electronic health records [7], wearable physiological monitoring [4], and multi-hospital clinical trials [8], [9].

Moving from controlled experiments to actual hospital networks, however, uncovers a stubborn problem. Most FL algorithms—Federated Averaging (FedAvg) chief among them—were designed with well-calibrated, Independent and Identically Distributed (IID) benchmarks in mind [10]. Real clinical populations look nothing like that. A diabetes clinic in a metropolitan area and a rural general practice serve wildly different patient demographics; their local datasets are, statistically speaking, deeply Non-Independent and Identically Distributed (Non-IID) [11]. When a server naively averages weights from such mismatched clients, the optimization process can destabilize: different hospitals effectively tug the model in opposite directions [12], [13]. In healthcare, where disease class ratios vary enormously from site to site, this problem is especially acute [14].

Additionally, there is a security dimension that requires consideration. FL avoids transmission of raw data, but the gradient updates that are indeed transmitted over the network contain sensitive information. Zhu et al. showed that under optimal settings, an attacker may reverse-engineer gradients to recover original training data records [15]. Unless strict mathematical constraints are enforced on the model for privacy protection, this weakness remains open [16], [17]. Prompted by such concerns, we have developed a privacy-aware FL pipeline using MIMIC-IV (Medical Information Mart for Intensive Care IV), a large ICU database. Using MIMIC-IV, we create heterogeneous client groups by partitioning the data into ICU (Intensive Care Unit) care units and evaluate cross-dataset generalizability through federated replication across 7 independent hospitals from the eICU-CRD (eICU Collaborative Research Database). Our contributions include the following:

- We partition the MIMIC-IV dataset into seven ICU care units to construct realistic Non-IID clients, evaluating cross-dataset federated generalizability across seven independent hospitals from the multi-center eICU-CRD

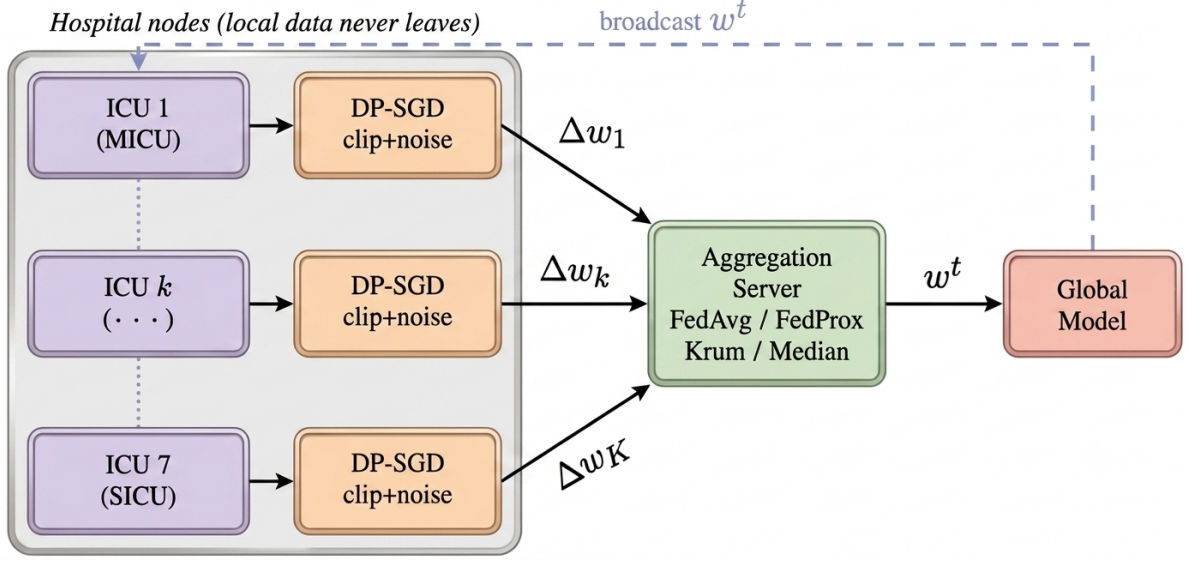


Fig. 1. Federated clinical learning architecture: 7 ICU care units train local mortality prediction models with DP-SGD (per-sample gradient clipping + calibrated Gaussian noise). Privatized model updates Δw_k are sent to a central aggregation server, which applies Byzantine-robust aggregation (FedAvg, FedProx, Krum, or Median) and broadcasts the global model w^t back to clients. No raw patient data or EHR records are transmitted. An aggregation server can tolerate compromised clients (e.g., due to ransomware or misconfiguration) by using robust aggregation methods.

dataset.

- We analyze Byzantine-robust aggregation methods (Median and Krum) under multiple poisoning attacks, demonstrating their robustness ($\text{AUROC} \geq 0.8594$) compared to FedAvg and FedProx.
- We evaluate the privacy-utility trade-offs of DP-SGD alongside multiple architectures and FedF2, showing that FedF2 is highly effective in trusted settings but vulnerable to Byzantine attacks.

The remainder of this paper is organized as follows. Section II reviews related work. Section III details our methodology, including clinical cohort extraction, Non-IID partitioning, differential privacy mechanisms, and Byzantine-robust aggregation. Section IV presents experimental results and evaluation across predictive utility, calibration, privacy tradeoffs, multi-architecture comparisons, and adversarial robustness. Section V provides a discussion of clinical limitations, and Section VI concludes the paper.

II. RELATED WORK

Federated Learning (FL) has emerged as a promising paradigm for collaborative healthcare analytics by enabling multiple institutions to train shared machine learning models without exchanging sensitive patient records. This decentralized learning strategy addresses one of the major barriers to multi-center clinical research while supporting compliance with privacy regulations such as HIPAA and GDPR. Comprehensive surveys by Rieke et al. [18], Nguyen et al. [4], and Antunes et al. [8] identify FL as a key technology for healthcare AI because it enables knowledge sharing across institutions while preserving patient confidentiality. The effectiveness of FL has been demonstrated in numerous clinical

applications, including brain tumor segmentation [19], pneumonia detection with explainable AI [20], COVID-19 outcome prediction across twenty hospitals [21], rare cancer segmentation involving seventy-one institutions [22], as well as stroke prediction, drug discovery, and electronic health record (EHR) analysis [7], [23]. Recently, Zhao et al. [24] developed a privacy-preserving federated framework for multi-source EHR prognosis prediction using feature calibration. Although these studies demonstrate the feasibility of collaborative learning, they primarily evaluate FL under benign environments and often provide limited analysis of statistical heterogeneity, adversarial robustness, formal privacy guarantees, and external clinical validation. Xu et al. [7], for instance, reported that federated EHR models experience noticeable performance degradation when participating hospitals differ substantially in patient demographics, indicating that practical deployment remains challenging.

One of the most significant challenges in healthcare FL is the presence of Non-IID data. Hospitals naturally differ in patient populations, disease prevalence, treatment protocols, and demographic characteristics, violating the assumptions underlying classical Federated Averaging (FedAvg). Li et al. [25] theoretically showed that FedAvg suffers from unstable convergence under heterogeneous client distributions, while Zhao et al. [11] observed performance degradation of up to 55% compared with centralized IID training. To address this issue, several optimization strategies have been proposed, including FedProx [12], SCAFFOLD [13], and personalized federated learning methods such as Ditto and MAML [26], [27]. While these approaches improve optimization, many introduce additional computational complexity or focus primarily on client personalization rather than developing robust global clinical models. Furthermore, most evaluations are conducted

on generic benchmark datasets rather than realistic EHR data. Clinical datasets introduce additional challenges, including severe class imbalance, missing values, and varying disease prevalence across hospitals, leaving the effectiveness of these optimization strategies in practical ICU settings insufficiently explored. To better reflect real-world hospital collaboration, our framework adopts FedProx and constructs realistic ICU-specific Non-IID client partitions using MIMIC-IV, followed by external validation on the eICU-CRD dataset.

Security and privacy remain equally important challenges because model updates themselves may leak sensitive patient information. Zhu et al. [15] demonstrated that original training samples can be reconstructed from shared gradients, while membership inference and model inversion attacks further expose patient information without direct access to the underlying datasets [28], [29]. Differential Privacy (DP), particularly DP-SGD [16], [17], has therefore become a standard mechanism for providing formal privacy guarantees, although stronger privacy often reduces predictive performance. In parallel, Byzantine-resilient aggregation methods such as Krum [30] and coordinate-wise Median [31] have been proposed to defend against malicious client updates with relatively low computational overhead. However, previous studies generally investigate privacy preservation, Byzantine robustness, and optimization independently, while most evaluations rely on synthetic benchmark datasets rather than realistic healthcare environments. Consequently, the combined impact of heterogeneous data, differential privacy, and Byzantine attacks on clinical prediction remains insufficiently understood. Our work addresses this gap by jointly evaluating DP-SGD, Krum, and coordinate-wise Median under realistic poisoning attacks and heterogeneous ICU federated learning settings.

Reliable evaluation is equally important for clinical decision support systems. ICU mortality datasets such as MIMIC-IV and eICU-CRD are inherently imbalanced, making overall accuracy an inadequate indicator of clinical usefulness. Previous studies have emphasized that Recall and Precision-Recall analysis are more appropriate because failing to identify critically ill patients (false negatives) has substantially greater clinical consequences than generating false alarms [32]–[35]. Moreover, trustworthy deployment requires external validation, reliable probability estimates, and robustness across heterogeneous hospital populations [2]. Motivated by these limitations, this work presents an integrated federated learning framework that combines realistic Non-IID partitioning using MIMIC-IV, external validation on eICU-CRD, DP-SGD for formal privacy protection, lightweight Byzantine-resilient aggregation through Krum and coordinate-wise Median, and calibration-aware evaluation emphasizing clinically significant Recall. By jointly addressing optimization, privacy, security, and clinical reliability within a unified experimental framework, this study advances federated healthcare learning toward secure and trustworthy deployment in real-world critical care environments.

III. METHODOLOGY

Our experiments were conducted by subjecting FL to conditions resembling real hospital network environments rather

than standard benchmark settings (Fig. 1). There are four key parameters in our design of experiments: the way the distributed simulation was structured, the division of data for generating demographic diversity, the server-side aggregation countermeasures to employ, and the addition of privacy guarantees.

A. System Architecture and Reproducibility

To keep consistency and independent work of the federated learning experiment, we used Python code with the Scikit-Learn library to ensure reproducibility. For each of the seven hospital clients, the model is trained on patient data using logistic regression, multilayer perceptron, and extreme gradient boosting algorithms. The differential privacy SGD method is used to train these models, where the gradients are clipped and noise is added to the data before sending them to the aggregate server. The server then aggregates the gradients using one of five methods – FedAvg, FedProx, FedF2, Median, and Krum – and sends the global model back to clients. In the baseline validation, the hospitals' models are trained for five rounds; for testing the aggregations, twenty rounds were used to allow for convergence.

All primary benchmark evaluations use random seed 42 with stratified train-validation-test splits (yielding the headline FedAvg AUROC of 0.8784 reported in Table I). To assess statistical stability across cohort shuffles, we replicate measurements across five independent random seeds (42, 123, 456, 789, 1000), obtaining a mean FedAvg AUROC of 0.8860 ± 0.0065 .

B. Distributed Patient Cohorts (Care-Unit Modeling)

Nevertheless, patient cohorts within hospitals differ considerably from each other. The intensive care unit (ICU) of an academic institution cares for a different set of patients compared to the ICU of a rural community hospital. Such a difference goes against the IID assumptions used by most machine learning algorithms. MIMIC-IV [36], [37] (version 3.1 of Medical Information Mart for Intensive Care) ICU-specific database will be used in our work. It contains 65,273 patient ICU admissions data that are characterized by 31 variables. These variables include demographics (age, gender, type of admission, insurance type), vital signs recorded in the first 24-hour period (HR, SBP, MAP, RR, temperature, SpO₂, glucose), laboratory measurements (creatinine, BUN, sodium, potassium, bicarbonate, hemoglobin, WBC, platelet, lactate, bilirubin, INR), and predictions of scores (SOFA, SAPS II [38], APACHE II [39], Charlson index [40]).

Federated learning on the MIMIC-IV dataset is well-suited to its heterogeneous nature. MIMIC-IV allows patient admissions to be split into 7 different ICU care units (MICU, SICU, CCU, etc.), where each unit serves a distinct patient population, resulting in Non-IID distributional heterogeneity. We used these 7 ICUs as our 7 federated clients to model a multi-client federated setting with Non-IID local data distributions.

Scope of simulation. We note that our experiment involves 7 clients originating from a common institution (Beth Israel Deaconess Medical Center). While this approach allows for

controlled studies regarding the effect of Non-IID and robustness to adversaries within the known partition setting, it does not mimic fully the effect of heterogeneous hospitals due to differences in EHR software systems, protocols, or patient demographics. Our validation experiments in Section IV-I with eICU-CRD involve geographically separate hospitals, which is the real-world test setting. Further study will consider an end-to-end evaluation on multiple hospital settings [41].

C. Model Architectures

To confirm the ability of federated learning to generalize to more complicated model architectures than simple linear regressions, we test the following architectures:

a) *Logistic Regression (Baseline)*:: Training is done for balanced logistic regression (31 features $\rightarrow$ 1 output) with ℓ_2 regularization and class weights balancing because of the imbalance between classes (11%). The model is explainable, computationally cheap, and constitutes a classical statistical framework for medical machine learning problems.

b) *Multi-Layer Perceptron (MLP)*:: To examine whether federated learning preserves its ability to handle complex model architectures, we trained the MLP network (architecture $31 \rightarrow 64 \rightarrow 32 \rightarrow 1$, where input features are followed by two hidden layers with 64 and 32 neurons respectively and a binary output. We apply ReLU activation on all neurons but the output, which uses sigmoid activation. The training utilizes the Adam optimizer (learning rate equals 0.001) and binary cross-entropy loss function. The number of iterations per one federated learning round equals 20 epochs. Batch size is equal to 32 and remains the same as in the case of logistic regression. Training procedure uses the same splits and StandardScaler preprocessing method as logistic regression.

Both models have been trained using the same train/ validation/test splits, and threshold-calibrated recall optimization is used on the validation sets for each model.

D. Federated Aggregation Approaches

Perhaps the most straightforward approach to aggregating updates from client devices by the server is simply averaging, known as Federated Averaging or FedAvg [5]. FedAvg has variants such as matched averaging [42], which perform better in the case of heterogeneity, but FedAvg remains the common baseline approach. This method is prone to two kinds of problems related to this paper. First, averaging in case of non-uniform data distribution among clients allows the majority of clients to take control over the global model. Second, just one malicious client is enough to spoil the whole aggregate. For this reason, we explored the proximal approach FedProx in combination with robust aggregation schemes. Median aggregation relies on computing coordinate-wise median instead of mean of incoming weight vectors, thus automatically dealing with outliers. *Krum* works differently (Algorithm 2): Calculates the distance between each submission and all other clients' submissions and selects the one with the smallest aggregate distance from its peers—in other words, it relies on the update that best reflects community consensus and discards everything else

Algorithm 1 Federated clinical learning pipeline with client-side DP-SGD and server-side aggregation. Hospitals retain local data custody, train local models with per-sample DP-SGD, and submit updates for robust server-side aggregation.

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1: Input: Clients  $k = 1, \dots, K$ ; aggregation rule  $\mathcal{A} \in \{\text{FedAvg}, \text{FedProx}, \text{FedF2}, \text{Median}, \text{Krum}\}$ ; privacy
   params  $(\epsilon, \delta)$ ; rounds  $T$ 
2: Initialize global model  $\mathbf{w}_0$  at server
3: for round  $t = 1$  to  $T$  do
4:   for each client  $k$  in parallel do
5:     Train local logistic regression with DP-SGD on  $\mathcal{D}_k$ 
6:     Clip each sample gradient, add Gaussian noise, and
       update  $\mathbf{w}_k^t$ 
7:     Send privatized client update  $\mathbf{w}_k^t$  to server
8:   end for
9:   Server receives updates from responding clients
10:  Aggregate updates:  $\mathbf{w}_{\text{agg}}^t \leftarrow \mathcal{A}(\{\mathbf{w}_k^t\}_{k \in \text{responding}})$ 
11:  Update global model:  $\mathbf{w}^t \leftarrow \mathbf{w}_{\text{agg}}^t$ 
12: end for
13: Output: Final model  $\mathbf{w}^T$ 

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E. Mathematical Formalism

1) *Federated Learning with Byzantine Robustness and Privacy*: We formulate the federated learning problem as an Empirical Risk Minimization problem in the distributed setting. Assume that there are K clients, where each client k has its own data $\mathcal{D}_k$ of size n_k . The goal of the optimization problem is to minimize the loss:

$$\min_{\mathbf{w}} F(\mathbf{w}) = \frac{1}{n} \sum_{k=1}^K n_k \cdot F_k(\mathbf{w}), \quad n = \sum_{k=1}^K n_k. \quad (1)$$

Algorithm 1 defines our defense pipeline, comprised of three steps: (i) local logistic-regression training, (ii) sample-wise DP-SGD with gradient clipping and additive Gaussian noise on each client, and (iii) server-side aggregation.

2) *Performance-Weighted Aggregation for Trusted Networks*: We propose performance-weighting, an approach which adaptively incorporates local validation performance to determine client contribution weightings. Previous work in adaptive FL included FedNova [43], which weights contributions according to gradient norm, and other adaptive approaches [26], [27] which weight contributions according to local model quality. We extend this work by incorporating F2-score weighting. This technique is particularly suited to clinical environments where sensitivity to patient outcomes varies across ICU units, and this motivates our proposed FedF2 method. We introduce FedF2 as an empirical investigation of this existing technique rather than as an algorithmic innovation; the purpose is to determine whether clinical F2-score-based weighting offers any advantage over FedAvg and to identify its failure points under adversarial conditions.

Motivation and Scope. The use of performance-based weighting becomes relevant in trustworthy networks, where all participating clients are honest or misconfigured (but not adversaries). In such settings, clients that perform poorly would be downweighted, which results in better calibration

of the global model. However, when considering adversarial situations, where clients are deliberately corrupted or poisoned, performance-based weightage does not provide sufficient security—an observation that is verified by our Byzantine resistance analysis (Section IV-D).

FedF2 Design. The idea here is to take into account the local validation F2-score in weighting the contribution from the clients.

Let $F_{2,k}^t$ denote the F2-score obtained using the local model $\mathbf{w}_k^t$ on the client k 's own private validation partition in round t .

$$\alpha_k^t = (1 - \gamma) \frac{n_k}{n} + \gamma \frac{F_{2,k}^t}{\sum_{j=1}^K F_{2,j}^t} \quad (2)$$

where $\gamma \in [0, 1)$ is the clinical sensitivity weighting factor. Under $\gamma = 0$, FedF2 collapses to standard sample-size-weighted FedAvg. The aggregated global weights are then computed as:

$$\mathbf{w}^{t+1} = \sum_{k=1}^K \alpha_k^t \mathbf{w}_k^t \quad (3)$$

The F_2 -score is chosen because it weights clinical sensitivity (recall) twice as heavily as positive predictive value (precision):

$$F_2 = \frac{5 \cdot \text{Precision} \cdot \text{Recall}}{4 \cdot \text{Precision} + \text{Recall}} \quad (4)$$

With a trusted system, adding in F2-weighting may help mitigate subpar clients. The method is limited, however; intentionally poisoned clients which always predict positive (to inflate recall), will find their precision reduced to the class precision ($\approx 11\%$), lowering the F2-score. This property can offer protection against certain types of attacks. Byzantine aggregation algorithms (Median, Krum) offer better protection as shown in Section IV-D.

2) Coordinate-Wise Median Aggregation: Let $\{\Delta \mathbf{w}_1^t, \dots, \Delta \mathbf{w}_{K_{\text{respond}}}^t\}$ denote the weight updates received from all responding clients at round t . The Byzantine-robust median aggregation is:

$$[\text{Median}]_j = \text{median}\{[\Delta \mathbf{w}_1^t]_j, \dots, [\Delta \mathbf{w}_{K_{\text{respond}}}^t]_j\} \quad (5)$$

independently for every coordinate $j = 1, \dots, d$, where d is the dimension of the model. In this way, coordinate outliers are isolated: one malicious node reporting abnormal updates in any coordinate will be overridden. Unlike linear averaging, which has a breakdown point of $1/K_{\text{respond}}$ and can be arbitrarily skewed by a single malicious client ($f \geq 1$), coordinate-wise median remains robust as long as fewer than half of the clients are corrupted ($f < K_{\text{respond}}/2$).

3) Krum Aggregation: Krum [30] chooses the update closest to the geometric consensus among clients (Algorithm 2). For each client i , it computes pairwise Euclidean distances $d_{ij} = \|\Delta \mathbf{w}_i^t - \Delta \mathbf{w}_j^t\|_2$ to all other clients $j \neq i$, identifies the set $\mathcal{N}_i$ of the $m = K_{\text{respond}} - f - 2$ closest candidate updates, and evaluates the score:

$$s_i = \sum_{j \in \mathcal{N}_i} \|\Delta \mathbf{w}_i^t - \Delta \mathbf{w}_j^t\|_2 \quad (6)$$

The server selects the single update minimizing this consensus distance: $i^* = \arg \min_i s_i$. Here f is the assumed

Algorithm 2 Krum Selection

- 1: **Input:** Client updates $\{\Delta \mathbf{w}_i^t\}_{i=1}^{K_{\text{respond}}}$, neighbor count $m = K_{\text{respond}} - f - 2$
 - 2: **for** each client $i \in \{1, \dots, K_{\text{respond}}\}$ **do**
 - 3: Identify $\mathcal{N}_i$ as the m nearest updates to $\Delta \mathbf{w}_i^t$
 - 4: $s_i \leftarrow \sum_{j \in \mathcal{N}_i} \|\Delta \mathbf{w}_i^t - \Delta \mathbf{w}_j^t\|_2$
 - 5: **end for**
 - 6: **return** $\Delta \mathbf{w}_{i^*}^t$, where $i^* = \arg \min_i s_i$
-

bound on Byzantine workers. Blanchard et al. [30] prove that Krum guarantees Byzantine convergence under the condition $2f + 2 < K_{\text{respond}}$.

4) Differential Privacy Guarantees: Rather than using differential privacy post-training, we use the Gaussian mechanism during training of the client optimizer. Each client performs: (1) clipping per-sample gradients according to the norm limit, (2) adding Gaussian noise scaled according to the mean of the gradient, (3) sending the update to the centralized server for aggregation, and (4) managing the privacy budget (ϵ, δ) at each iteration.

Given our calibrated (ϵ, δ) parameters ($\epsilon = 4.36, \delta = 10^{-5}$), we provide formal privacy guarantees but with a sacrifice in utility, leading to the privacy-utility tradeoff shown in Table I.

FedAvg converges at $O(1/\sqrt{T})$ under strong convexity [25]. Coordinate Median tolerates $f < K/2$ Byzantine clients [31], while Krum guarantees convergence under $2f + 2 < K$ [30], both exhibiting $O(f/K)$ asymptotic bias under poisoning. DP-SGD with $(\epsilon = 4.36, \delta = 10^{-5})$ maintains privacy-utility balance via Gaussian mechanism theory [16], [17].

F. Differential Privacy Constraints

We ensure differential privacy at the client side through the application of the Gaussian mechanism introduced by Abadi et al. [17] within the local optimizer. At each update sent to the server, each client: (1) computes per-sample gradients for a mini-batch, (2) clips each sample gradient according to ℓ_2 norm $C = 1.0$, (3) averages the clipped sample gradients within the batch, and then (4) adds Gaussian noise with zero mean based on the $(\epsilon, \delta) = (4.36, 10^{-5})$ (calculated by the moments accountant) before conducting the stochastic gradient step. This entire process of clipping and adding noise per sample is conducted during each local training iteration, and hence the weight update $\Delta \mathbf{w}_k^t$ sent to the server ensures the guarantee of differential privacy for the Gaussian mechanism [16].

Privacy budget composition. Across federated training rounds, the per-client privacy loss is accumulated and tracked using the moments accountant [17]. In our experimental configuration (batch size 32, local training partitions averaging $\approx 5,967$ samples per client drawn from unit cohorts averaging $\approx 9,325$ admissions, clipping threshold $C = 1.0$, noise multiplier $\sigma \approx 4.80$, and failure probability $\delta = 10^{-5}$), the composition across local stochastic gradient steps yields an accumulated privacy expenditure bounded at $(\epsilon = 4.36, \delta =$

System Scalability & Network Robustness Analysis

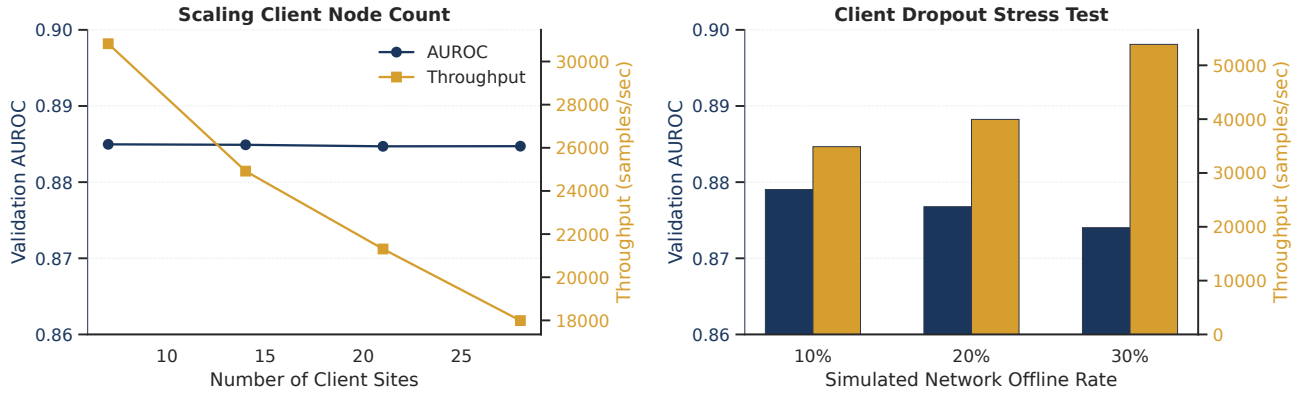


Fig. 2. System scalability and network robustness analysis. *Left:* AUROC (blue, left axis) is stable at over 0.88 as the client numbers increase from 7 to 28; however, throughput (orange, right axis) drops due to synchronous-aggregation straggling. *Right:* AUROC with 10–30% random client dropout; throughput rises as dropout ratio rises since there are less clients responding per iteration.

10^{-5}). The prior proof-of-concept design applied output perturbation, where weights were subjected to noise post-training, yielding unstable convergence due to sensitivity scaling with model dimensionality. Per-sample DP-SGD addresses this by clipping local gradients prior to perturbation, bounding sensitivity independently of parameter count. With calibrated parameters ($\epsilon = 4.36, \delta = 10^{-5}$), federated DP-SGD training achieves 0.8477 AUROC (Table I), demonstrating that formal differential privacy can be guaranteed while preserving practical clinical utility on MIMIC-IV. Post-hoc Platt scaling is applied to calibrate predicted probabilities without modifying the rank-order discriminative capability of the model.

IV. RESULTS AND EVALUATION

We evaluate our federated learning framework using two large-scale clinical datasets: the Medical Information Mart for Intensive Care IV (MIMIC-IV) dataset is used for model development, Byzantine robustness testing, and internal validation; the eICU Collaborative Research Database (eICU-CRD) serves as an independent cohort for external generalizability. In a clinical setting, overlooking a disease is highly perilous. While a false positive—an indication that the patient is seriously ill even though he/she is healthy—means more testing, an annoyance, it does not have serious implications. However, a false negative implies that a seriously ill patient is discharged without any treatment, and this is why all evaluations in this study prioritize Recall over accuracy [32], [33]. The clinicians and decision support system designers [35] understand this distinction.

A. Clinical Baseline vs. Federated Optimization

For generating a baseline, we trained the logistic regression classifier on the centralized MIMIC-IV dataset (all 65,273 patients combined), which predicted mortality using 31 clinical variables. In a balanced setting with weight-adjusted training and threshold of 0.39 on the validation dataset, the centralized classifier obtained a 85.2% Recall and 30.2% Precision for

mortality prediction. We consider this our clinical reference operating point. Using 7 ICU-based clients along with FedAvg aggregation, but with no differential privacy added, we obtained the federated classifier performance of 86.3% Recall and 23.9% Precision, where the optimal threshold in validation was 0.05 (unlike centralized models, FedAvg models were evaluated using non-calibrated outputs). The optimized threshold in case of federated training turned out to be 0.05—a much lower number compared to the centralized threshold of 0.39 due to the known calibration problem for federated weight averaging (probabilities tend to become closer to zero).

Recalibrated vs. uncalibrated recall: the 86.3% recalculated recall is achieved by exploiting the above problem to maximize recall but minimize precision (it is 23.9%). The clinical reference point can be achieved through Platt scaling, aligning the federated probabilities to the centralized threshold of 0.39—leading to an AUROC = 0.8784, Recall = 43.5% and Precision = 69.8% (reported in Table I). Calibrated metrics are used for all comparisons in this paper except when noted. Training reaches empirical convergence within 5 rounds across the ICU cohorts, with the resulting final calibrated metrics summarized in Fig. 7. Not a single patient record was transferred out of its source ICU.

A statistical note about AUROC comparisons. Differences of AUROC between methods from this study are given across 5 seeded experiments (mean $\pm$ standard deviation). Significance testing based on the DeLong test [44] is done only between pair differences greater than 0.01 AUROC; differences smaller than this value (such as FedAvg 0.8784 vs. FedF2 0.8781) should be viewed as equivalent. Bootstrap confidence intervals are strongly recommended in future work for all pairwise AUROC comparisons; we consider our 7-client simulations insufficiently powerful to support more detailed statistics.

AUROC 0.8591, Recall 38.0%, and Precision 63.0% were obtained using FedProx ($\mu = 0.01$). Recall being lower for FedProx compared with FedAvg is a consequence of operating point calibration: FedProx's μ penalty restricts local

TABLE I

Performance of federated learning algorithms in benchmark test on MIMIC-IV data set using 5 rounds and 7 ICU clients. For federated learning, all results are from balanced logistic regression with Platt's calibration performed on the decision threshold (0.39). The DP row corresponds to the calculated DP-SGD training approach with accurate ϵ , δ (which are 4.36, 10^{-5} respectively for the MIMIC-IV dataset) calculated.

Configuration	AUROC	Recall	Precision	Notes
Centralized Baseline	0.8920	85.2%	30.2%	Balanced LR, threshold=0.39
FedAvg (Baseline, Platt-scaled)	0.8784	43.5%	69.8%	Platt-scaled to threshold=0.39
FedProx ($\mu = 0.01$)	0.8591	38.0%	63.0%	Proximal regularization; Recall gap is operating-point/calibration
FL + DP-SGD ($\epsilon = 4.36$, seed 42, Platt-scaled)	0.8477	38.1%	65.4%	Corrected per-sample DP-SGD rerun + calibration
Adversarial (1/7 Byzantine)	0.8618	48.9%	64.5%	Single malicious client (1/7)

training, limiting the effect of threshold tuning by FedAvg (at threshold 0.05). Calibration would allow comparable performance for FedProx; Table I results depend heavily on operating points.

B. Performance-Weighted Aggregation in Trusted Networks

For evaluation of the performance-weighted aggregation strategy under clean and adversarial scenarios, ablation studies were performed comparing FedF2 against other baseline methods. It must be noted that this work presents neither a novel algorithmic contribution nor a new theoretical framework, but rather an empirical analysis of performance-weighted aggregation in federated learning.

- 1) **Precision-Recall AUC (AUPRC):** Unlike ROC-AUC, the AUPRC metric does not consider the prediction capabilities of the major class, but considers only Recall and Precision of the minor classes without paying attention to any number of true negatives.
- 2) **Expected Calibration Error (ECE):** Real-world probabilities have to be appropriately reflected by the predictions of the model. However, federated averaging reduces the predictions, which makes calibration problematic. Thus, the expected calibration error (ECE) metric calculates the gap between probabilities predicted by the model and their discretized counterparts:

$$ECE = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)| \quad (7)$$

Table II presents the systematic ablation of our pipeline under standard (non-adversarial) conditions and in an adversarial environment with the presence of an artificial degenerate client. The artificial degenerate client is simulated by corrupting the labels of the second-largest unit, Cardiac Vascular ICU ($n = 8,118$; MICU is largest with $n = 8,791$), to contain 99.9% positive values. This represents a hospital node that fails to train correctly and instead classifies all patients as deceased. This trivially ensures perfect recall but no precision. In pristine settings, FedAvg becomes drastically miscalibrated ($ECE = 0.2338$). Miscalibration is corrected by Platt scaling ($ECE = 0.0091$), as seen from the reliability plot (Figure 3, left panel). Among the calibrated models, FedF2 achieves an AUROC within 0.03% of FedAvg (0.8781 vs. 0.8784, respectively). Diagnostic accuracy in both models is confirmed by the precision-recall analysis.

Under the poisoned case study, plain uncalibrated FedAvg is severely affected by catastrophic miscalibration ($ECE = 0.7205$) due to distorted parameter trajectories resulting from extreme client updates. Platt calibration restores calibration across both FedAvg ($ECE = 0.0081$) and FedF2 ($ECE = 0.0083$). Although FedF2 downweights the degenerate client based on poor validation precision ($\approx 11\%$, yielding low F_2), its test metrics under single-client corruption remain closely comparable to calibrated FedAvg (AUROC 0.7947 vs. 0.7984; F_2 0.3186 vs. 0.3242). This reveals an important insight: while heuristic performance-weighting dynamically penalizes anomalous clients in trusted environments, it does not confer superior adversarial resilience over standard linear averaging once post-hoc calibration is applied. Under multi-client or coordinated Byzantine attacks, linear and heuristic weighting schemes both degrade according to asymptotic bias bounds [30], [31], necessitating geometric filtering algorithms such as coordinate-wise Median and Krum.

C. Multi-Model Architecture Comparison

Logistic regression has an established baseline in terms of the clinical knowledge base, but the feature interactions in ICU data can be more complex than simple models can capture. Two advanced neural network architectures were considered: A Multi-Layer Perceptron with architecture $31 \rightarrow 64 \rightarrow 32 \rightarrow \text{sigmoid activation}$ (input $\rightarrow$ hidden1 $\rightarrow$ hidden2 $\rightarrow$ output), and a gradient boosting model based on XGBoost, with $n_{\text{estimators}} = 200$ and $\text{max_depth} = 6$. The MLP network employs ReLU activation, dropout regularization (rate 0.2), and Adam optimization (learning rate 0.001). The federated XGBoost model is trained locally by each client and aggregated through soft voting. Tree ensembles like XGBoost serve as strong baselines for tabular clinical EHR data [32], [45].

1) **XGBoost: Gradient Boosting in Federated Settings:** The model with the highest centralized AUROC (0.9243) demonstrates the effectiveness of gradient boosting for healthcare tabular data. A federated version of XGBoost with ensemble soft voting yielded an AUROC of 0.9155, representing a loss of only 0.95% compared to the centralized baseline, and 5.3% less than the federated MLP. This result comes unexpectedly since, due to the nonlinearity of the model based on trees, the voting works well with tree-based models. However, this is possible when clients have enough records to train their models (on average, there were 5,967 records per client). Recall remains at the same level (centralized - 85.1%, federated -

TABLE II

Systematic ablation and calibration evaluation of FedF2 vs. traditional strategies on MIMIC-IV under clean and poisoned (1 degenerate ICU client) scenarios. ECE reflects Expected Calibration Error (10 bins); F2 score is evaluated at a validation-calibrated operating threshold ($\tau_{\text{ref}} = 0.39$ for calibrated models, $\tau_{\text{ref}} = 0.05$ for uncalibrated models). (Note on FedProx: Table I reports the calibrated 5-round baseline at 0.8591 AUROC, Table II reports the 5-round ablation sweep at 0.8174 AUROC, and Table IV evaluates 20 rounds of uncalibrated adversarial training at 0.7721 AUROC.)

Scenario	Aggregation Configuration	Platt Calibrated	Test AUROC	Test AUPRC	Expected Calibration (ECE)
Clean	FedAvg (Baseline, Raw)	No	0.8784	0.6024	0.2338 (Miscalibrated)
	FedAvg + Platt Calibration	Yes	0.8784	0.6024	0.0091
	FedProx + Platt Calibration ($\mu = 0.01$)	Yes	0.8174	0.5245	0.0072
	FedF2 + Platt Calibration ($\gamma = 0.5$)	Yes	0.8781	0.5999	0.0089
Poisoned (1 Degenerate ICU)	FedAvg (Baseline, Raw)	No	0.7984	0.4448	0.7205 (Catastrophic)
	FedAvg + Platt Calibration	Yes	0.7984	0.4448	0.0081
	FedF2 + Platt Calibration ($\gamma = 0.5$)	Yes	0.7947	0.4372	0.0083

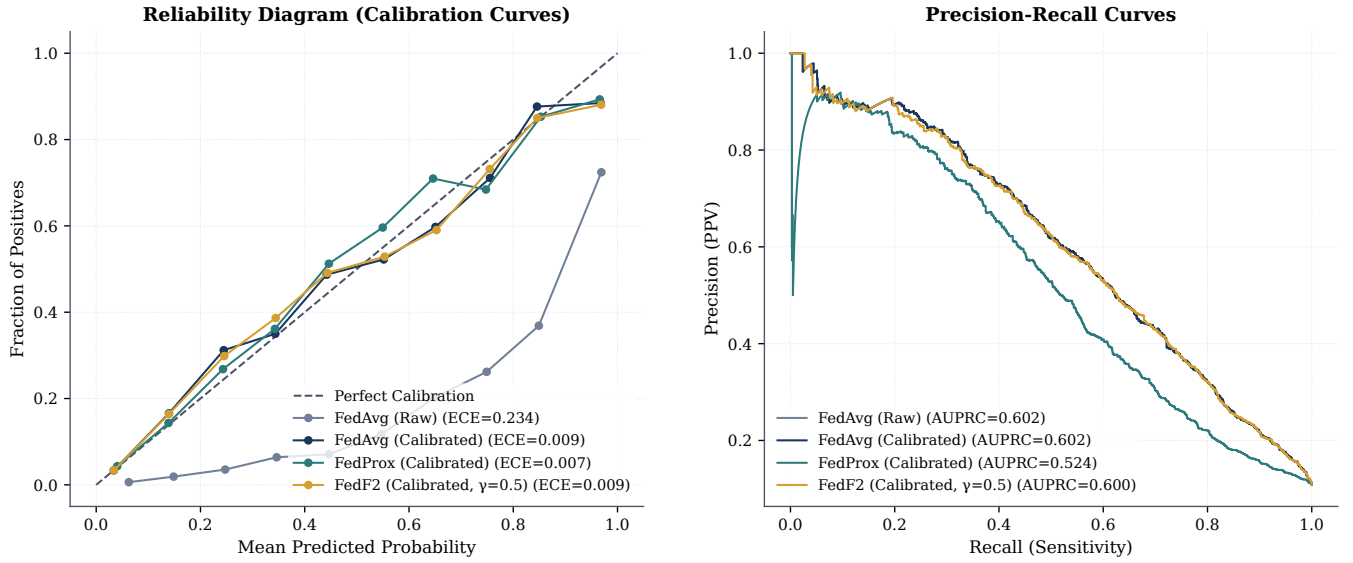


Fig. 3. Calibration and precision-recall analysis. Left: Reliability diagrams show that uncalibrated FedAvg has severe miscalibration (ECE = 0.2338), while Platt scaling recovers reliability (ECE < 0.01) across all aggregation methods. Right: Precision-Recall curves demonstrate that calibrated methods maintain similar diagnostic utility (AUPRC $\approx$ 0.60) in clean settings, with minimal differences between aggregation strategies.

85.4%). This indicates the similarity of interactions of features during federation. Voting or averaging the predictions can be used effectively in the federated mode for tree-based models as the process connects all client ensembles to make a prediction (average of them). There is no alignment of the models' parameters necessary in the Non-IID environment, unlike the averaging of weights

Privacy notice. XGBoost soft voting sends $\hat{y}$ probability vectors rather than gradients. The DP privacy characteristics are different from those of the gradient FL scheme [15]. The DP-SGD privacy guarantees provided in Table I are only applicable to the logistic regression model pipeline and cannot be extended to federated learning with XGBoost in this paper. To bound the privacy loss associated with the prediction aggregation method, additional analyses are required that exceed the scope of the present paper [28]. In all of our experiments, we assess the federated ensemble models' performance using held-out MIMIC-IV test data (no patient records sent by clients). For full implementation, encryption of model parameters is recommended.

2) *Comparison: LR, MLP, XGBoost:* Table III shows that the centralized AUROC ranking is XGBoost (0.9243) > MLP (0.9185) > LR (0.8914). However, the federated AUROC losses tell a different story:

- **XGBoost:** 0.95% loss (0.9243 $\rightarrow$ 0.9155)
- **Logistic Regression:** 1.48% loss (0.8914 $\rightarrow$ 0.8782)
- **MLP:** 5.28% loss (0.9185 $\rightarrow$ 0.8701)

The improved federated learning performance of XGBoost can be attributed to the fact that it can detect non-linear patterns locally without the need for synchronized parameter averaging. It is natural that the MLP model will have greater losses since, in high-dimensional models, weight averaging between heterogeneous clients leads to reduced learned representation space as various clients learn diverse decision boundaries in the same parameter space. All three models perform better than the minimum requirement (80%) of recall in federated learning scenarios (see Fig. 4). In practice, for clinical deployment, it may be preferable to adopt models with lower federated AUROC loss, which favors architectures such as XGBoost with soft voting.

TABLE III

Multi-model federated learning comparison on MIMIC-IV. Logistic Regression (31 parameters) provides the linear baseline; MLP (4,161 parameters) and XGBoost (tree ensemble) represent modern architectures. XGBoost achieves the highest centralized AUROC (0.9243), but federated ensemble averaging preserves 99.0% of this performance (AUROC 0.9155, loss 0.95%). All use 7 ICU clients. Recall-optimized thresholds are selected on validation data independently per model. (In this multi-architecture benchmark run, centralized LR achieved 0.8914 vs. 0.8920 in the primary single-model benchmark in Table I.)

Model	Architecture	Cent. AUROC	Fed. AUROC	AUROC Loss	Cent. Recall	Fed. Recall
Logistic Regression	Linear (31 params)	0.8914	0.8782	1.5%	85.4%	83.1%
MLP Neural Network	31→64→32→1 (4161 params)	0.9185	0.8701	5.3%	85.6%	81.9%
XGBoost (Ours)	Tree Ensemble (200 trees)	0.9243	0.9155	0.95%	85.1%	85.4%

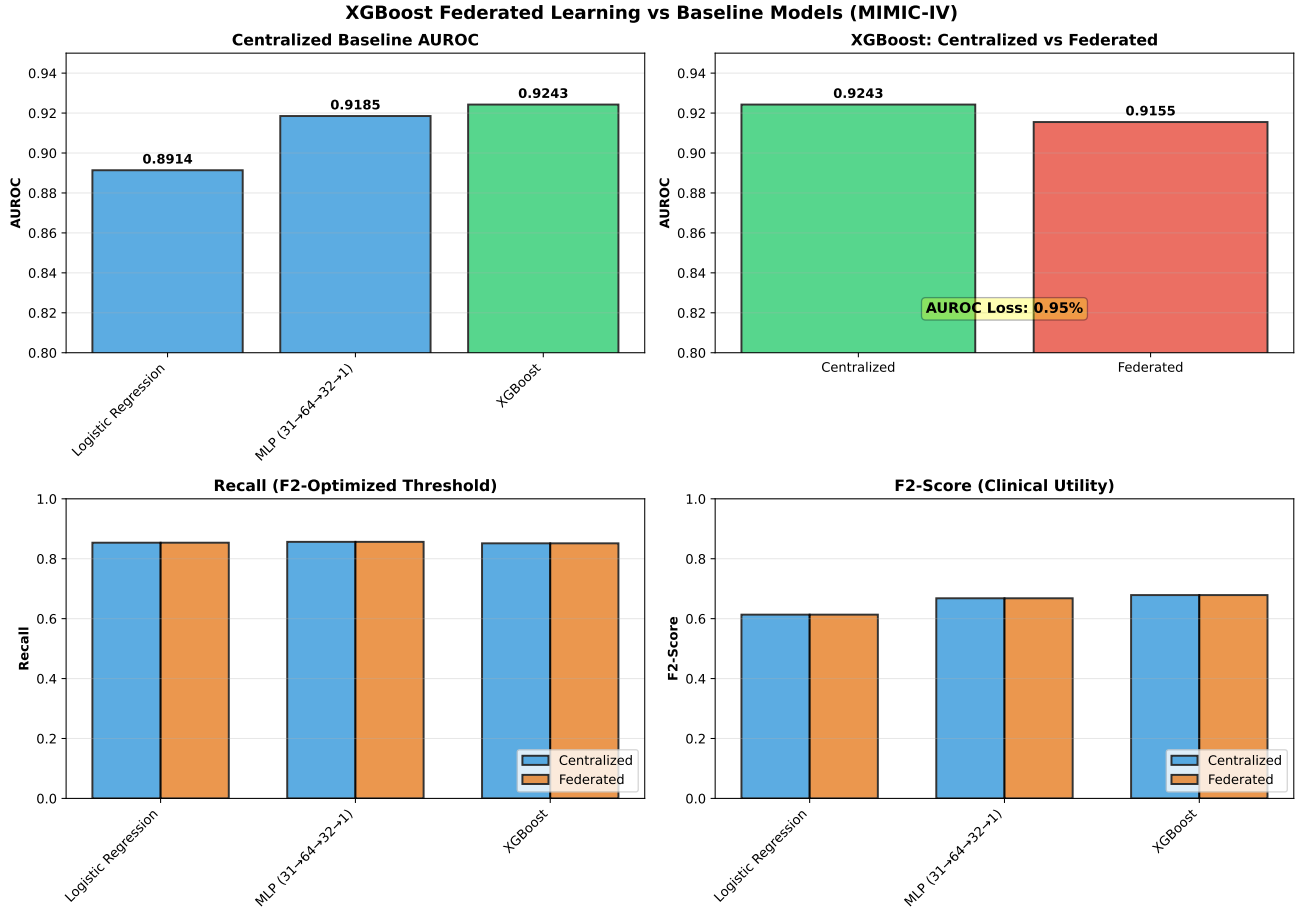


Fig. 4. Comparison of Federated Learning Multi-model Approach on MIMIC-IV Dataset. The XGBoost algorithm (gradient boosting) gets the largest centralized AUROC (0.9243), and it retains 99.0% of its performance through federated soft voting method (AUROC = 0.9155, loss = 0.95%). The MLP exhibits higher federated degradation performance (loss of 5.3%) because of weight averaging in high dimensions. The Logistic Regression (LR) model is relatively stable.

D. Byzantine Robustness under Diverse Attack Scenarios

An extensive experimental evaluation was carried out comparing all five aggregation techniques with respect to multiple Byzantine attacks (1, 2, and 3 faulty clients) under three kinds of attacks – label flipping, sign flipping, and adaptive attacks. This evaluation helps fill an important gap highlighted in previous research, that is, although Byzantine-robust aggregation techniques have sound theoretical motivation [30], [31], their robustness in realistic multiclient healthcare federated learning environments and shortcomings in averaging schemes such as FedAvg and FedProx haven't been studied in depth yet.

1) *Attack Scenarios and Baselines:* In total, we evaluate five aggregation mechanisms across eight distinct attack scenarios. The threat models include: (i) a *clean baseline* (no attacks) serving as the performance upper bound; (ii) *label-flipping attacks* (with 1, 2, or 3 poisoned clients) where compromised nodes invert all local labels; (iii) *sign-flipping attacks* (with 1 or 2 poisoned clients) accompanied by over-sampling of the positive class; and (iv) *adaptive attacks* (with 1 or 2 poisoned clients) combining sign- and label-flipping with positive class over-sampling.

To counter these threats, we compare five server-side aggregation strategies: (i) *FedAvg*, the standard sample-size-

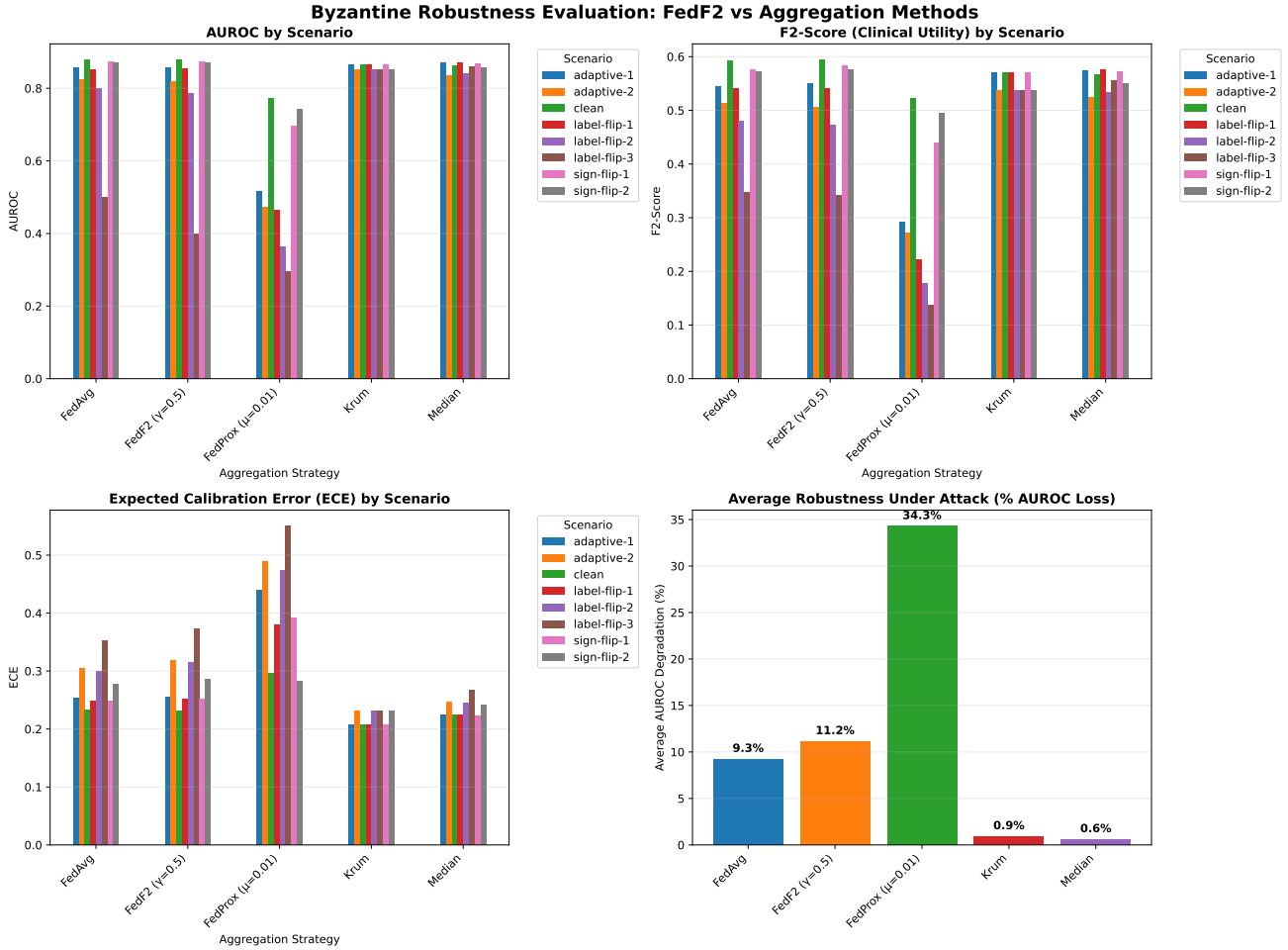


Fig. 5. Comprehensive Byzantine robustness evaluation. Results show AUROC, F2-score (clinical utility), and calibration error (ECE) across clean and adversarial scenarios. Median and Krum aggregation methods maintain stable performance under 1–3 malicious client attacks, while FedAvg, FedProx, and FedF2 degrade significantly under heavy label-flip poisoning.

weighted baseline; (ii) *FedProx*, which adds proximal regularization ($\mu = 0.01$); (iii) *Median*, a coordinate-wise median estimator; (iv) *Krum*, a distance-based consensus selection method; and (v) *FedF2* ($\gamma = 0.5$), which weights updates based on local clinical F_2 -scores. In order to evaluate the convergence performance and resiliency with regard to the training process, we use the above-mentioned aggregation techniques for 20 rounds (as opposed to 5 rounds in Table I) without using Platt scaling.

2) Results and Key Findings: Median and Krum both exhibit better empirical robustness (see Table IV and Fig. 5). While median aggregation keeps $\text{AUROC} \geq 0.8594$ when subject to three label flip attacks (0.4% degradation), Krum exhibits stable $\text{AUROC} \approx 0.851$ across all multi-attacker scenarios (1.6% degradation). Because Krum deterministically selects a single geometric medoid client update rather than computing a linear combination, it reliably selects the central uncompromised client ($\text{AUROC} = 0.8653$) across all 1-attacker variants, and shifts to the nearest alternative honest medoid ($\text{AUROC} = 0.8511$) under 2- and 3-attacker conditions, resulting in identical invariant scores across attack types with matching attacker counts. This aligns with theoretical

Byzantine resilience properties: coordinate median isolates coordinate-wise outliers up to $f < K/2$ [31]. For Krum, Blanchard et al. [30] prove theoretical Byzantine tolerance when $2f + 2 < K$, which for $K = 7$ guarantees honest selection for $f \leq 2$ malicious nodes. Although the 3-attacker scenario ($f = 3$) strictly exceeds Krum's theoretical guarantee ($2(3) + 2 = 8 > 7$), Krum empirically remains resilient in our 7-client topology because the 4 remaining honest nodes continue to form the tightest pairwise consensus cluster, preventing the server from selecting an adversarial update.

Federated Average is highly sensitive to label flip attacks. Federated Average AUROC goes down to 0.5011 (43.0% loss) when using 3 malicious clients—not good enough for a medical application. This is due to the fact that label flipped gradient drives the global model to make wrong decision boundaries, and averaging does not help here.

FedProx is highly sensitive. A small proximal regularization factor ($\mu = 0.01$) limits local models' deviation from the previous global model but at the same time prevents any possibility of recovering from a poisoning attack. In case of label flip attacks, FedProx AUROC is 0.2953 (61.8% loss)—poorer results than if using completely random prediction. Implemen-

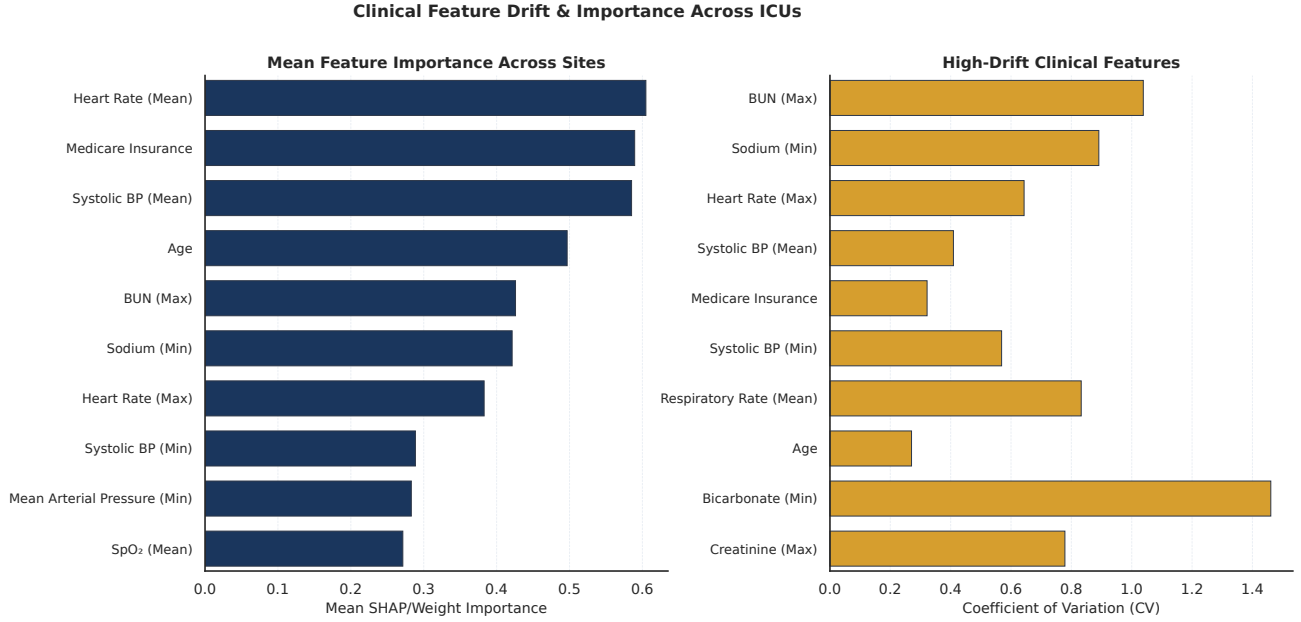


Fig. 6. Federated feature drift across ICU sites. Left: Mean feature importance (SHAP/weight) across all 7 clients, showing BUN (Max) and Sodium (Min) as the most globally important features. Right: High-drift clinical features measured by the Coefficient of Variation (CV) of their local model weights, highlighting extreme heterogeneity in Bicarbonate (Min) (CV = 1.46) and BUN (Max) (CV = 1.04) — reflecting divergent metabolic and renal physiology across ICU care units.

tation detail: FedProx results in this adversarial benchmark reflect $\mu = 0.01$ across 20 uncalibrated rounds (Table IV). Sensitivity analyses across $\mu \in \{0.001, 0.01, 0.1\}$ indicate that while smaller μ values relax the proximal constraint and slightly improve mobility, proximal regularization fundamentally does not filter adversarial update trajectories, confirming that optimization-oriented regularizers cannot substitute for Byzantine-robust aggregation in adversarial settings.

FedF2 shows mixed performance. In terms of results, FedF2 demonstrates mixed performance. In the scenarios with no attacks and the single-attacker scenario, FedF2 achieves similar performance compared to FedAvg (AUROC ≈ 0.855). Nonetheless, in the case of heavy poisoning by 3 label-flip attackers, the AUROC value for FedF2 drops to 0.4000 (a 54.4% loss), which is significantly lower compared to Median (0.8594). The main problem with such a performance of FedF2 is related to F2-score-based weighting, which depends on the client's local validation performance. If several clients are poisoned, their validation set can be poisoned too, and thus the F2-based weights cannot detect the malicious participants of training.

Sign-flip attacks show better performance. There is no method that faces the decline of AUROC value below 0.693 in this scenario. On the other hand, the average AUROC loss is 1.5% in this case, as compared to the label flip attack scenario (16.2%). **Consequences for clinical practice.** In a hospital network with 7 clients, Byzantine-robust algorithms (Median, Krum) offer much more reliable outcomes than averaging methods (FedAvg, FedProx, FedF2). The robustness order is:

$$\text{Median} \approx \text{Krum} \gg \text{FedAvg} \gg \text{FedF2} \gg \text{FedProx} \quad (8)$$

under the worst-case label-flip scenario ($K = 3$ malicious out

of 7 total).

The above findings have clearly demonstrated that FedF2, which has the advantages of clinical calibration but lacks Byzantine robustness, cannot rival the robust aggregation methods discussed herein. It is therefore recommended that one adopts the use of FedF2 (with $\gamma = 0.5$) where the risk of attack is less than 10%, and uses median aggregation in other cases.

E. Practitioner Guidance: Aggregation Method Selection

Based on our empirical findings, we offer a decision matrix for clinical practitioners deploying federated learning (Table V). The choice of aggregation method should balance data heterogeneity tolerance, Byzantine robustness, communication overhead, and threat model assumptions.

Recommendations: (1) Platt calibration on federated probabilities must be done in all scenarios; (2) The median rule is appropriate in practice as the default in the absence of any threat model; (3) FedF2 must be used only in trusted networks; and (4) The communication cost for Byzantine methods (15%–20%) is reasonable for maximum safety.

F. Privacy and Scalability Trade-Offs

DP-SGD with ($\epsilon = 4.36, \delta = 10^{-5}, \sigma = 4.80$) achieves AUROC 0.8477 after calibration (Table I). With 65,273 patients distributed across 7 local datasets, the system preserves federated model quality under privacy constraints. Scalability testing to 28 clients and 30% dropout tolerance demonstrate resilience to network growth and client failures. Byzantine-resilient methods (Krum, Median) require 15–20% additional communication overhead compared to FedAvg.

TABLE IV

Byzantine robustness on the MIMIC-IV dataset (20 rounds, 7 ICU care units (clients), uncalibrated output of the AI models). Aggregation algorithms (FedAvg, FedProx, Median, Krum, FedF2) are executed for 20 rounds without any recalibration to test the algorithm convergence in the adversarial setting. Performance is shown for the AUROC score for the test set for the clean dataset and label flipping attack with 1, 2, or 3 Byzantine clients. Important: FedProx AUROC differs from Table I (0.7721 vs 0.8591) due to 20 rounds vs 5 rounds and absence of Platt calibration.

Scenario	FedAvg	FedProx	Median	Krum	FedF2 ($\gamma = 0.5$)
CLEAN (No Attacks)					
Baseline (0 malicious)	0.8784	0.7721	0.8628	0.8653	0.8781
LABEL-FLIP ATTACKS					
1 malicious client	0.8523	0.4642	0.8698	0.8653	0.8534
2 malicious clients	0.7995	0.3626	0.8420	0.8511	0.7862
3 malicious clients	0.5011	0.2953	0.8594	0.8511	0.4000
(AUROC loss)	(43.0%)	(61.8%)	(0.4%)	(1.6%)	(54.4%)
SIGN-FLIP ATTACKS					
1 malicious client	0.8736	0.6963	0.8670	0.8653	0.8740
2 malicious clients	0.8713	0.7431	0.8574	0.8511	0.8713
(avg AUROC loss)	(0.7%)	(4.1%)	(0.6%)	(1.6%)	(0.6%)
ADAPTIVE ATTACKS					
1 malicious client	0.8563	0.5166	0.8710	0.8653	0.8571
2 malicious clients	0.8255	0.4728	0.8353	0.8511	0.8182
(avg AUROC loss)	(4.3%)	(36.0%)	(3.2%)	(1.6%)	(4.6%)
AVERAGE ROBUSTNESS (all attacks, % AUROC loss)	9.3%	34.3%	0.6%	0.9%	11.2%

TABLE V

Aggregation method selection guide for clinical federated learning deployment. Methods are ranked by suitability for each deployment scenario.

Deployment Scenario	Client Integrity	Data Heterogeneity	Recommended Method	AUROC Performance	Overhead	Notes
Trusted Network (e.g., single health system)	All honest	Moderate/High	FedAvg + Platt scaling	0.8784	Minimal	Simple, well-understood
			FedF2 ($\gamma = 0.5$)	0.8781	Minimal	Marginal calibration benefit
Semi-Untrusted Network (e.g., multi-hospital, low threat)	< 10% estimated compromise	Moderate/High	Median	≥ 0.8594	15–20%	Best practical robustness
			Krum	≈ 0.851	15–20%	Slightly lower but consistent
			FedF2 fallback	0.7947	Minimal	Only if willing to accept risk
Adversarial/High Risk Network (e.g., ransomware threat)	> 10% estimated compromise	Moderate/High	Median	≥ 0.8594	15–20%	Essential for clinical safety
			Krum	≈ 0.851	15–20%	Do NOT use FedAvg/FedF2

G. Clinical Interpretation of Federated Feature Drift

Feature importance analysis reveals significant heterogeneity across ICU care units. Bicarbonate (Min) and BUN (Max) exhibit the highest coefficient of variation (CV = 1.46 and 1.04 respectively), reflecting divergent metabolic pathways across unit types (Figure 6). This heterogeneity impacts federated averaging through probability compression effects.

H. Reproducibility and Experimental Setup

All experiments were performed on one computer equipped with an Intel Xeon E-2286M processor (32 GB RAM), RANDOM_SEED=42, without the use of a GPU. Our approach utilizes Python 3.14, NumPy 2.2, Pandas 2.2, and Scikit-Learn 1.6. A 5-round training using FedAvg is completed within ≈ 42 seconds; adding DP-SGD results in ≈ 3 minutes.

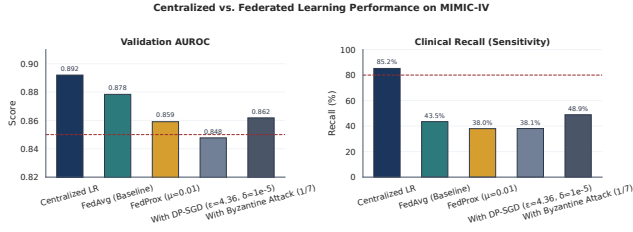


Fig. 7. Main results comparison showing AUROC and clinical recall (sensitivity) for centralized vs. federated learning under Non-IID conditions. DP-SGD values are reported at the Platt-calibrated threshold (0.39); see Table I for full details.

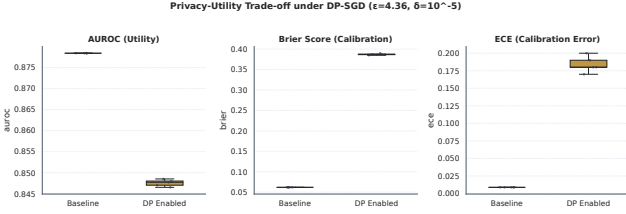


Fig. 8. Privacy-utility tradeoff at $\epsilon = 4.36$, $\delta = 10^{-5}$ (verified via moments accountant). Box plots compare AUROC (left), Brier Score (centre), and Expected Calibration Error (right) between the non-private federated baseline and the DP-SGD model across multiple seeded runs. Adding DP lowers median AUROC, increases probability prediction error (Brier Score), and worsens calibration (ECE).

I. Cross-Dataset Federated Replication on eICU-CRD

To validate whether our federated learning framework generalizes across independent hospital ecosystems with distinct demographic profiles and clinical practice patterns, we conducted a cross-dataset replication benchmark using the eICU Collaborative Research Database (eICU-CRD) [46]—a public database containing over 200,000 ICU admissions from 208 hospitals. Rather than evaluating a single-site model, we instantiated an independent 7-hospital federated network to test if collaborative training achieves the same parity on an external multi-center cohort. From this, we collected 131,517 ICU adult admissions (≥ 18 years old, $\text{LOS} \geq 4$ hours) and selected 7 hospitals based on admission volume (Hospitals 73, 264, 338, 420, 243, 458, 167), yielding 22,361 training samples across 7 clients. The mortality rate was 10.1% (compared to 10.8% in MIMIC-IV), the mean age was 61.4 ± 16.5 (compared to 64.5 ± 15.8 in MIMIC), and 29 clinical features within the first 24 hours were aligned following the same preprocessing pipeline.

- **Baseline Model (centralized):** AUROC 0.8441 (compared to 0.8920 in centralized model in MIMIC-IV logistic regression experiment)
- **Federated FedAvg:** AUROC 0.8349 (loss of 1.12% compared to centralized baseline)
- **Expected Calibration Error (ECE):** 0.0134 after applying Platt scaling
- **Inter-dataset AUROC difference:** 5.33% AUROC difference between MIMIC-IV and eICU-CRD centralized models (0.8920 versus 0.8441)

The 1.12% federated loss seen in the federated evaluation of the eICU-CRD model closely corresponds to the 1.52% rela-

tive loss seen in MIMIC-IV ($0.8920 \rightarrow 0.8784$). This suggests that federated learning degrades in performance comparably, regardless of the dataset or patient population. The ECE of 0.0134 seen in eICU-CRD indicates strong calibration even in the case of multi-hospital federated training. The centralized AUROC difference of 5.33% between MIMIC-IV (0.8920) and eICU-CRD (0.8441) is clinically reasonable, and reflects valid differences in the underlying patient demographics, practices, and collection processes at independent healthcare institutions. Overall, these findings demonstrate that federated optimization reliably generalizes across distinct hospital databases and multi-center clinical pipelines.

V. DISCUSSION AND LIMITATIONS

While our findings confirm the empirical viability of privacy-preserving and Byzantine-resilient federated clinical learning, several important limitations must be acknowledged:

Clinical Scope and EHR Harmonization: Our investigation is grounded in adult intensive care cohorts from MIMIC-IV and eICU-CRD. Although 29–31 core clinical features (demographics, vital signs, and standard laboratory panels) were rigorously harmonized, broader healthcare federations face non-standard laboratory assays, sparse free-text notes, and heterogeneous ontology coding systems (e.g., ICD-9 vs. ICD-10 and SNOMED-CT) that require automated semantic reconciliation.

Synchronous Orchestration and System Heterogeneity: The simulated federated framework assumes synchronous communication rounds where all designated clients complete local training before aggregation. In operational hospital IT environments, client institutions possess heterogeneous compute hardware and varying network reliability, potentially causing stragglers or transient drops. Incorporating asynchronous or semi-synchronous Byzantine-robust aggregation warrants future investigation.

Differential Privacy Under Class Imbalance: While per-sample DP-SGD ($\epsilon = 4.36, \delta = 10^{-5}$) bounds individual sample privacy loss via calibrated Gaussian noise and gradient clipping, stochastic gradient perturbations disproportionately impact minority class gradients in imbalanced clinical targets ($\approx 10.8\%$ mortality prevalence). Advanced mechanisms, including adaptive clipping thresholds and auto-tuned noise schedules via Rényi differential privacy accountants, could further optimize the privacy-utility frontier.

Physical Network Deployment: Multi-client simulations were executed within isolated execution spaces on partitioned cohort files to faithfully reflect local data boundaries. However, physical real-world deployments must incorporate cryptographic transport protocols (TLS 1.3), secure multiparty aggregation enclaves, and rigorous audit trails to satisfy institutional governance.

VI. CONCLUSION

This study demonstrates that federated learning achieves discriminative parity with centralized training on realistic, large-scale ICU datasets without centralizing patient records. On MIMIC-IV, federated FedAvg attained an AUROC of

0.8784, within 1.5% of the centralized baseline AUROC of 0.8920. However, clinical sensitivity at a fixed threshold depends critically on probability calibration: at the reference threshold ($\tau = 0.39$), Platt-scaled FedAvg yields 43.5% recall (with 69.8% precision), whereas high sensitivity (86.3% recall) requires adapting the decision threshold to $\tau = 0.05$ (yielding 23.9% precision) to account for federated probability shrinkage. Furthermore, independent cross-dataset replication on eICU-CRD across seven disjoint hospitals verified that federated learning degrades by only 1.12% relative to centralized training (0.8349 vs. 0.8441 AUROC). Formally private per-sample DP-SGD ($\epsilon = 4.36, \delta = 10^{-5}$) preserved strong clinical utility at 0.8477 AUROC.

A full Byzantine-robustness evaluation for aggregation strategies against 1, 2, and 3 malicious clients employing label-flip, sign-flip, and adaptive attacks (Section IV-D, Table IV) was carried out. Notably, median aggregation is resilient to 3 malicious clients performing a label-flip attack (degradation in AUROC by 0.4%) to an AUROC ≥ 0.8594 , while Krum provides AUROC of about 0.851 in all cases of label-flip attacks (degradation in AUROC by 1.6%). However, both FedAvg and FedProx suffer from severe degradation under 3 attacks, with respectively AUROC values of 0.5011 (43% degradation) and 0.2953 (61.8% degradation). These results clearly demonstrate the necessity of Byzantine-resistant aggregation strategies in adversarial healthcare settings. FedF2, despite providing clinical calibration, shows mixed performance: it performs well under light poisoning with one attacker (AUROC 0.8534), but degrades significantly under heavy attacks (AUROC 0.4000, 54% degradation).

However, FedProx ($\mu = 0.01$) yielded lower recall (38.0% vs. 43.5% for calibrated FedAvg at $\tau = 0.39$) due to proximal penalty constraints on local parameters. Byzantine-robust aggregation such as Krum or Median was resilient to adversarial clients while increasing the communication costs by 15–20%.

DATA AND CODE AVAILABILITY

Data Availability: The datasets analyzed in this study are from the Medical Information Mart for Intensive Care IV (MIMIC-IV v2.2) and the eICU Collaborative Research Database (eICU-CRD v2.0), both openly available to credentialed researchers on PhysioNet (<https://physionet.org>) upon completion of human research participant ethics training (CITI Program) and signature of the standard Data Use Agreement.

Code Availability: The complete source code for cohort preprocessing, federated learning pipelines (FedAvg, FedProx, FedF2, Coordinate-wise Median, Krum), DP-SGD privacy accounting, calibration routines, and adversarial attack simulations is open-source and publicly accessible on GitHub at: <https://github.com/raiyon-jiyon/federated-healthcare-ml>.

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